Understanding Generalization and Memorization of Deep Neural Network

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Definition

Memorization

- Overfitting to outliers, random labels, or noise.

Generalization Gap

- Difference between train performance and test performance.
- In general, memorization leads to a large generalization gap (not always).

Contents

Understanding Deep Learning requires rethinking Generalization (citation: 5700, NeurIPS 2017).

- Modern neural networks are already too complex that it can memorize the all training dataset.
- Therefore, it opens up research questions why DNNs generalize, not memorize.

A Closer Look at Memorization in Deep Networks (citation: 1600, ICLR 2017).

- When the dataset contains simple patterns, DNNs tend to generalize.
- In the absence of such patterns, DNNs are more likely to memorize.

What Neural Networks Memorize and Why: Discovering the Long Tail (Zhang et al., NeurIPS 2020)

- Memorization is not always BAD.
- Memorization improves generalization as real-world dataset is long-tailed.

Understanding Deep Learning requires rethinking Generalization

Key Ideas

1. From the perspective of traditional learning theory, the model's complexity is defined by the prediction accuracy of the model, trained from the random labels.
   - The authors found that the modern DNNs can easily trained to predict the random labels.
   - Hard to understand why DNNs generalize not memorize.
2. Does the generalization stems from regularization objectives? (constraining function classes?)
   - No. DNNs can still generalize without any regularization objectives.
3. Then why does neural network generalize so well?
   - Modern DNN architectures with gradient descent may prefer simpler function classes.

Understanding Deep Learning requires rethinking Generalization

Traditional Complexity Measure: Rademacher Complexity

- Commonly used metric to measure the complexity of the hypothesis class (=model).
- Measures how precisely the model can predict the random labels.

hypothesis class $\mathcal{H}$ on a dataset $\{x_1, \ldots, x_n\}$ is defined as

$$\hat{\mathfrak{R}}_n(\mathcal{H}) = \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} \sigma_i h(x_i) \right]$$

where $\sigma_1, \ldots, \sigma_n \in \{\pm 1\}$ are i.i.d. uniform random variables.

- In theory, a smaller Rademacher complexity indicates a smaller model complexity.

Understanding Deep Learning requires rethinking Generalization

Randomization Experiment

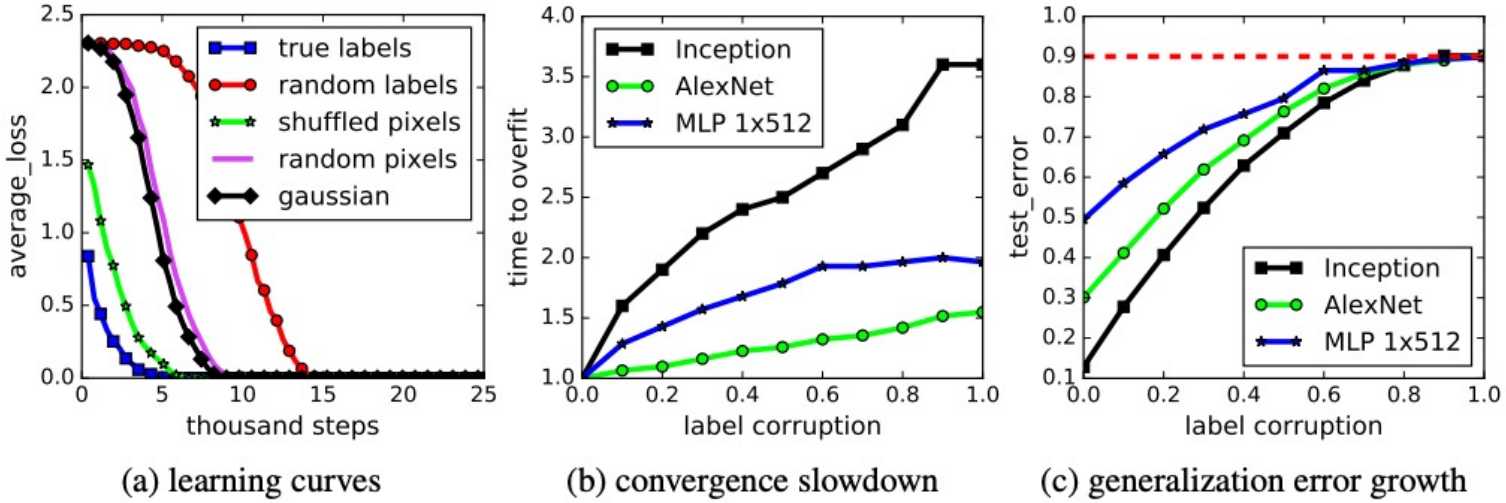


Figure 1: Fitting random labels and random pixels on CIFAR10. (a) shows the training loss of various experiment settings decaying with the training steps. (b) shows the relative convergence time with different label corruption ratio. (c) shows the test error (also the generalization error since training error is 0) under different label corruptions.

- Although the convergence takes a bit longer, the model can easily fit the random labels.

Understanding Deep Learning requires rethinking Generalization

Randomization Experiment & Role of Regularization

model	# params	random crop	weight decay	train accuracy	test accuracy
Inception	1,649,402	yes	yes	100.0	89.05
		yes	no	100.0	89.31
		no	yes	100.0	86.03
		no	no	100.0	85.75
(fitting random labels)		no	no	100.0	9.78
Inception w/o BatchNorm	1,649,402	no	yes	100.0	83.00
		no	no	100.0	82.00
(fitting random labels)		no	no	100.0	10.12
Alexnet	1,387,786	yes	yes	99.90	81.22
		yes	no	99.82	79.66
		no	yes	100.0	77.36
		no	no	100.0	76.07
(fitting random labels)		no	no	99.82	9.86
MLP 3x512	1,735,178	no	yes	100.0	53.35
		no	no	100.0	52.39
(fitting random labels)		no	no	100.0	10.48
MLP 1x512	1,209,866	no	yes	99.80	50.39
		no	no	100.0	50.51
(fitting random labels)		no	no	99.34	10.61

- Regardless of the architecture, deep neural network can easily fit the random labels (Same Rademacher Complexity).
- Even without any regularization, DNNs can generalize well (Generalization is not primarily driven by Regularizations).

Understanding Deep Learning requires rethinking Generalization

Implicit Regularization of Gradient Descent

- Let's assume we initialize the initial weight to 0. Then, which solution does SGD converges to?

to. Since the SGD update takes the form $w_{t+1} = w_t - \eta_t e_t x_{i_t}$ where η_t is the step size and e_t is the prediction error loss. If $w_0 = 0$, we must have that the solution has the form $w = \sum_{i=1}^{n} \alpha_i x_i$ for some coefficients α. Hence, if we run SGD we have that $w = X^T \alpha$ lies in the span of the data points. If we also perfectly interpolate the labels we have $Xw = y$. Enforcing both of these identities, this reduces to the single equation

$$XX^T \alpha = y \tag{3}$$

We can therefore perfectly fit any set of labels by forming the Gram matrix (aka the *kernel matrix*) on the data $K = XX^T$ and solving the linear system $K\alpha = y$ for α. This is an $n \times n$ linear system that can be solved on standard workstations whenever n is less than a hundred thousand, as is the case for small benchmarks like CIFAR10 and MNIST.

Quite surprisingly, fitting the training labels exactly yields excellent performance for convex models. On MNIST with no preprocessing, we are able to achieve a test error of 1.2% by simply solving (3).

- Solving the linear equation of the gram matrix converges to a unique solution with good generalization.

Understanding Deep Learning requires rethinking Generalization

Summary

1. Deep Neural Networks tend to Generalize not Memorize.
2. Traditional Learning Theory or Regularization methods cannot explain this phenomenon.
3. Implicit Regularization effect of Gradient Descent provides one viable explanation for generalization.

A Closer Look at Memorization in Deep Networks

Key Ideas

1. There exist different patterns of DNN optimization on real data vs noise.

   DNNs generalize on real data while DNNs memorize on random data.

2. DNNs Learn simple patterns first, before memorizing.
   - Connects to the result of [1].
   - The network converges to the low rank and then gradually increase its rank.

[1] The Low-Rank Simplicity Bias in Deep Networks, Huh et al, TMLR 2022.

A Closer Look at Memorization in Deep Networks

DNNs generalize on real data while memorize on random data

- Hypothesis
  - Let's assume DNNs generalize on real data while DNNS memorize on random data.
  - Then, on real data, there exist "easy examples" that fit these patterns better than the others.
  - However, on random data, all the examples would be equally difficult.

A Closer Look at Memorization in Deep Networks

DNNs generalize on real data while memorize on random data

- Existence of Easy Examples (CIFAR-10, 3-Conv layers, 100seeds).

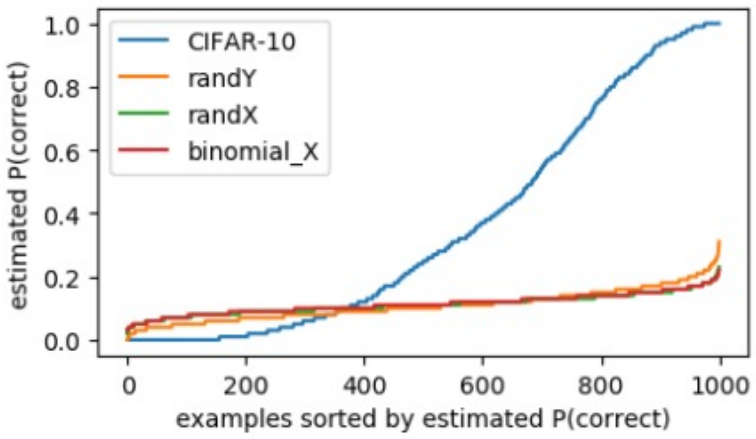
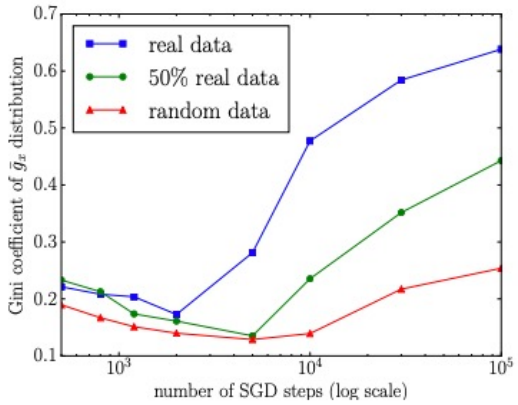


Figure 1. Average (over 100 experiments) misclassification rate for each of 1000 examples after one epoch of training. This measure of an example's difficulty is much more variable in real data. We conjecture this is because the easier examples are explained by some simple patterns, which are reliably learned within the first epoch of training. We include 1000 points samples from a binomial distribution with $n = 100$ and p equal to the average estimated P(correct) for randX, and note that this curve closely resembles the randX curve, suggesting that random inputs are all equally difficult.

- Loss Sensitivity in Real vs Random Data.



- Real: High Gini-coefficient of gradient w.r.t each example (unfair).
- Random: Low Gini-coefficient of gradient w.r.t each example (fair).

A Closer Look at Memorization in Deep Networks

DNNs learn simple patterns first, then memorize.

- Memorization naturally leads to complex decision boundary.
- How can we measure the complexity of decision boundary?
  - Critical Sample
    - A sample is defined to be critical, if the model predicts different label on nearby data point.

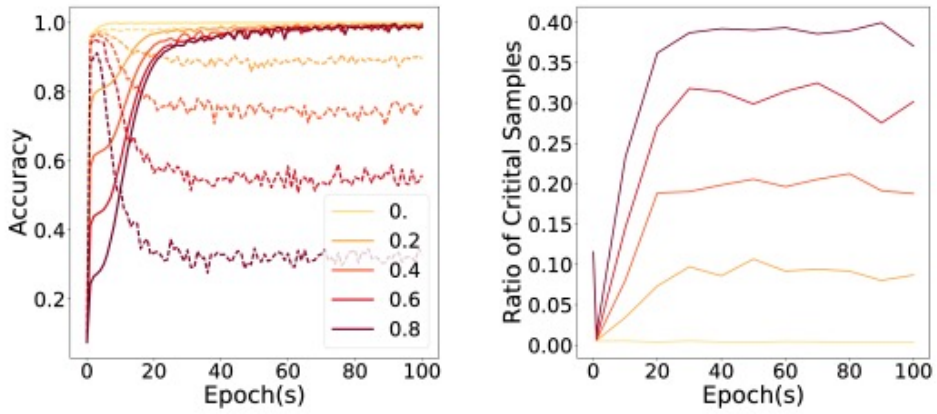
$$\arg\max_i f_i(\mathbf{x}) \neq \arg\max_j f_j(\hat{\mathbf{x}})$$
$$\text{s.t. } \|\mathbf{x} - \hat{\mathbf{x}}\|_\infty \leq r$$

  - Critical Sample Ratio (CSR)
    - Given a model and dataset, CSR represents a fraction of critical samples in a dataset.
    - High CSR: Complex decision boundary (≈ Memorization).
    - Low CSR: Simple decision boundary (≈ Generalization).

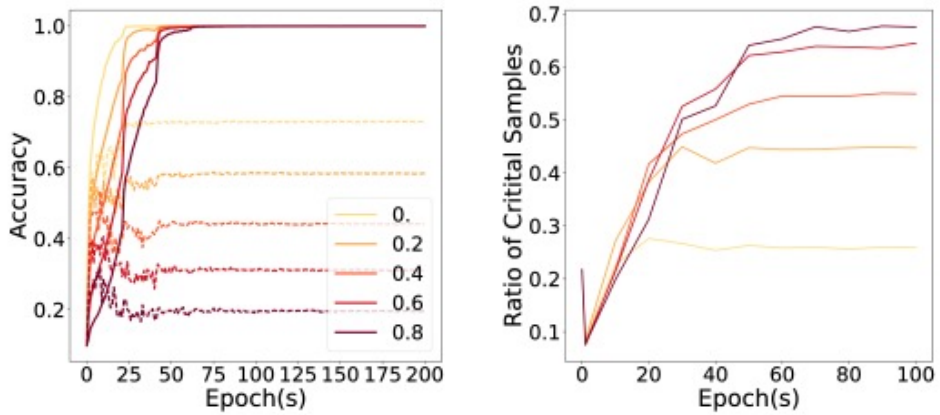
A Closer Look at Memorization in Deep Networks

DNNs learn simple patterns first, then memorize.

- Dataset: MNIST (Top) and CIFAR-10 (Bottom).



(b) Noise added on classification labels.



(b) Noise added on classification labels.

- During the course of training, CSR decreases then increases (generalize then memorize).
- With higher label noise, memorization leads to higher CSR and lower validation accuracy.

A Closer Look at Memorization in Deep Networks

Summary

1. DNNs generalize on real data while DNNs memorize on random data.
2. DNNs first generalize then memorize (Highest validation accuracy then gradually decreases).
3. Memorization is critical when trained with random labels.

What Neural Networks Memorize and Why

Key Ideas

1. Define the measure to quantify the example-level memorization.
2. Without memorized examples, DNNs exhibit poor generalization performance.
3. This is because real-world datasets are long-tailed.

[1] The Low-Rank Simplicity Bias in Deep Networks, Huh et al, TMLR 2022.

What Neural Networks Memorize and Why

Quantifying example-level memorization

Example-level "Generalization Gap"

$$\frac{1}{|\mathcal{F}_z^\bullet|} \sum_{f \in \mathcal{F}_z^\bullet} P(f, z) - \frac{1}{|\mathcal{F}_z^\circ|} \sum_{f' \in \mathcal{F}_z^\circ} P(f', z)$$

avg perf. of models f that have seen eg z

avg perf. of models f' that have not seen eg z

Given an example z

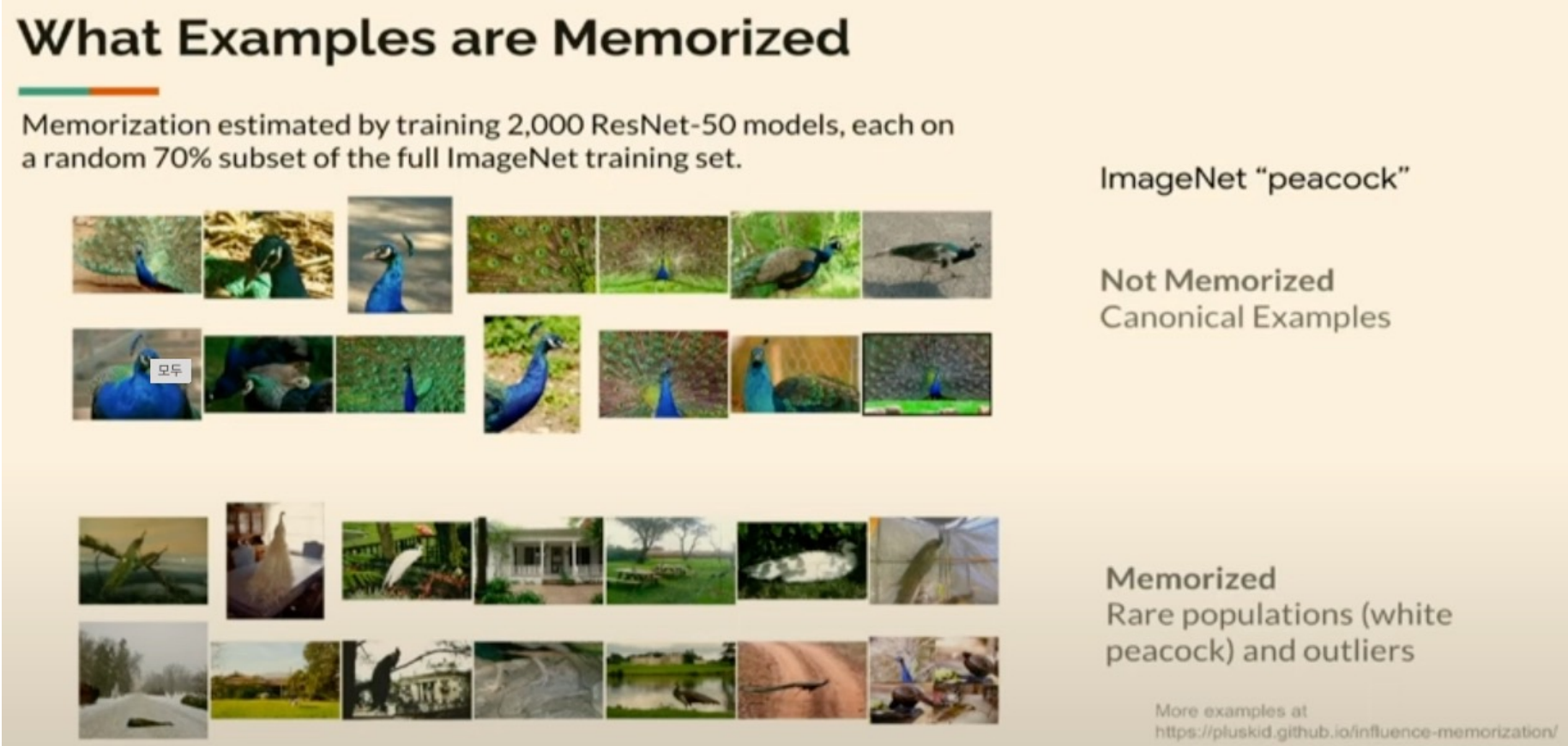
1. Train a number of models with z (IN models)
2. Train a number of models without z (OUT models)
3. Compute the average performance of IN models on z minus the average performance of OUT models on z

We call the example z "memorized" if this "generalization gap" is large.

[1] Quantifying and Understanding Memorization in Deep Neural Networks, Talk from Chiyuan Zhang, 2023.

What Neural Networks Memorize and Why

What Examples are Memorized? (=Large Generalization gap).



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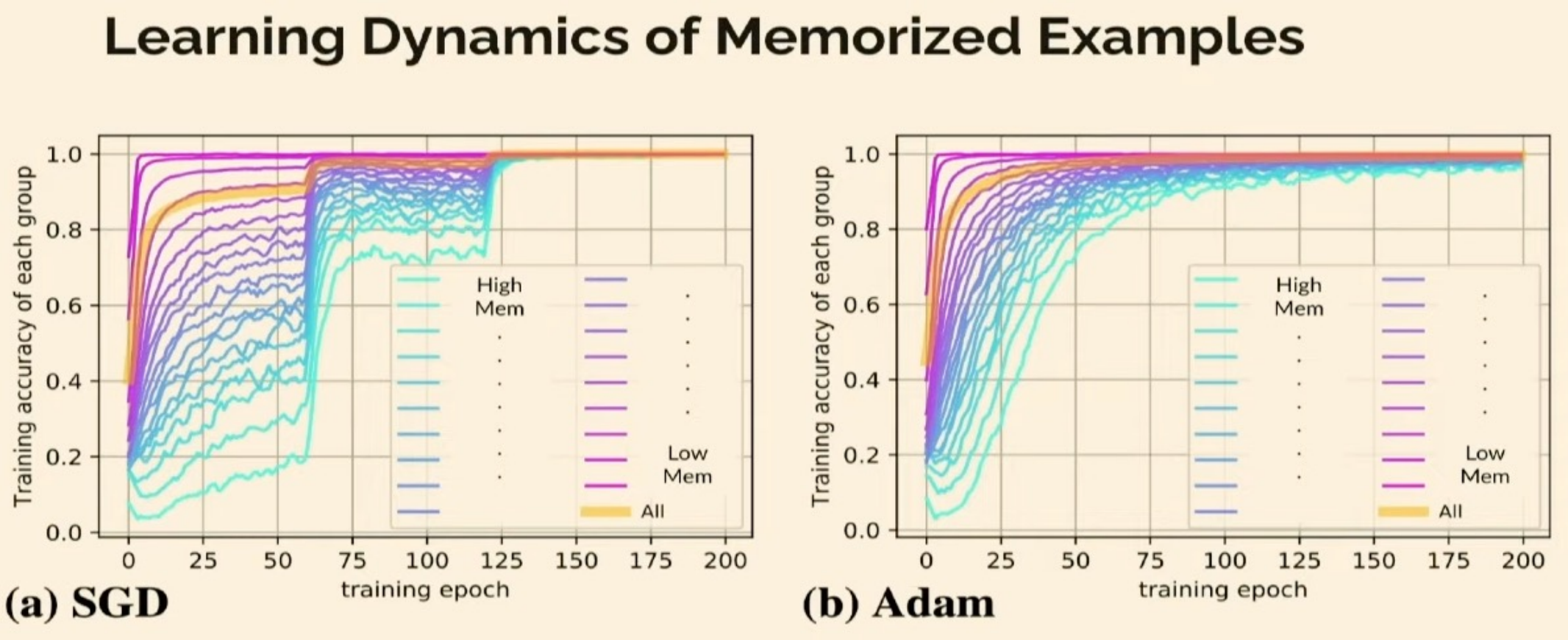
What Neural Networks Memorize and Why

Learning Dynamics of Memorized Samples.



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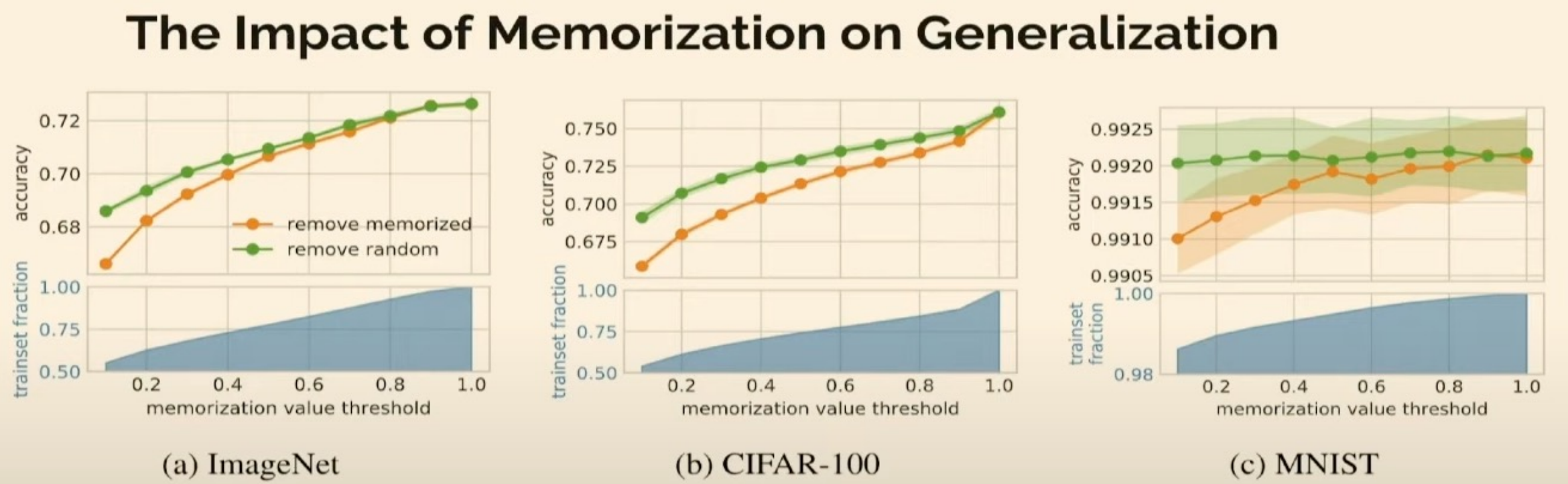
What Neural Networks Memorize and Why

The Impact of Memorization on Generalization



But in many cases, removing memorized examples **hurt** generalization; and it **hurts more** than removing random examples.

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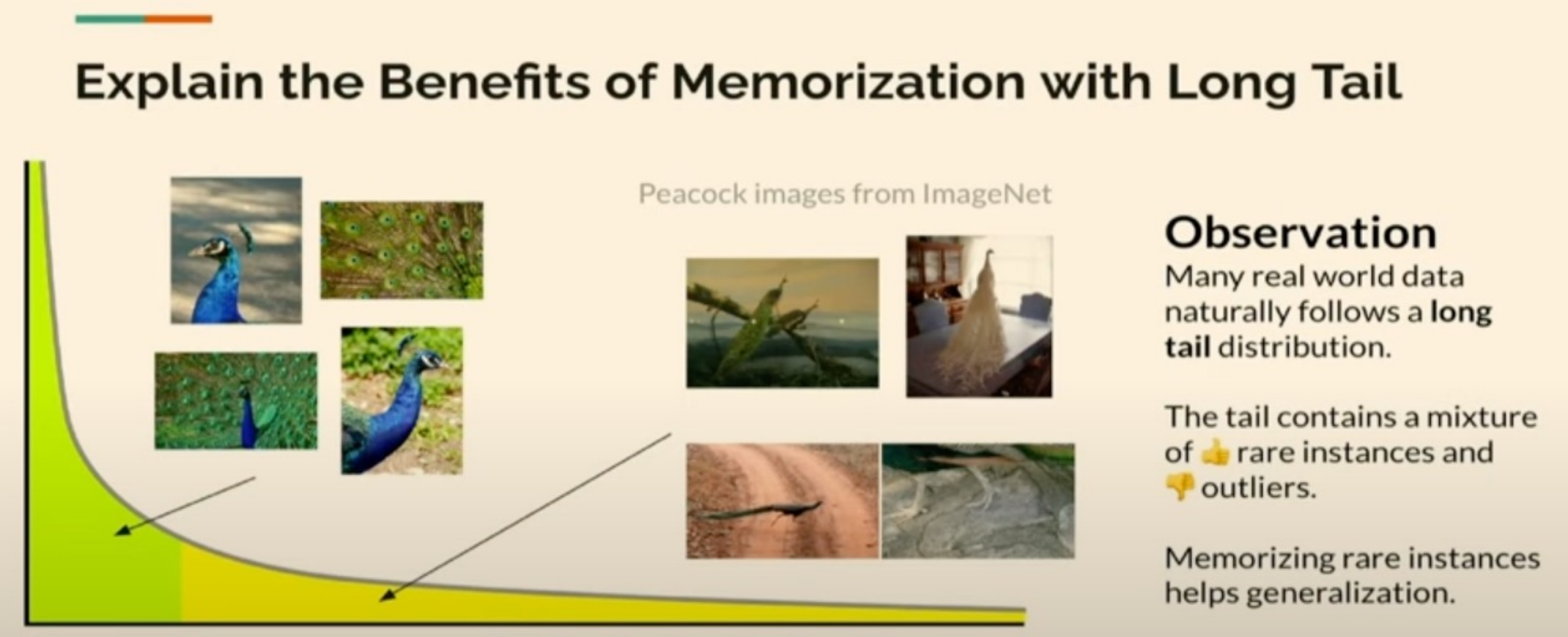
What Neural Networks Memorize and Why

Why does Memorization improves Generalization?



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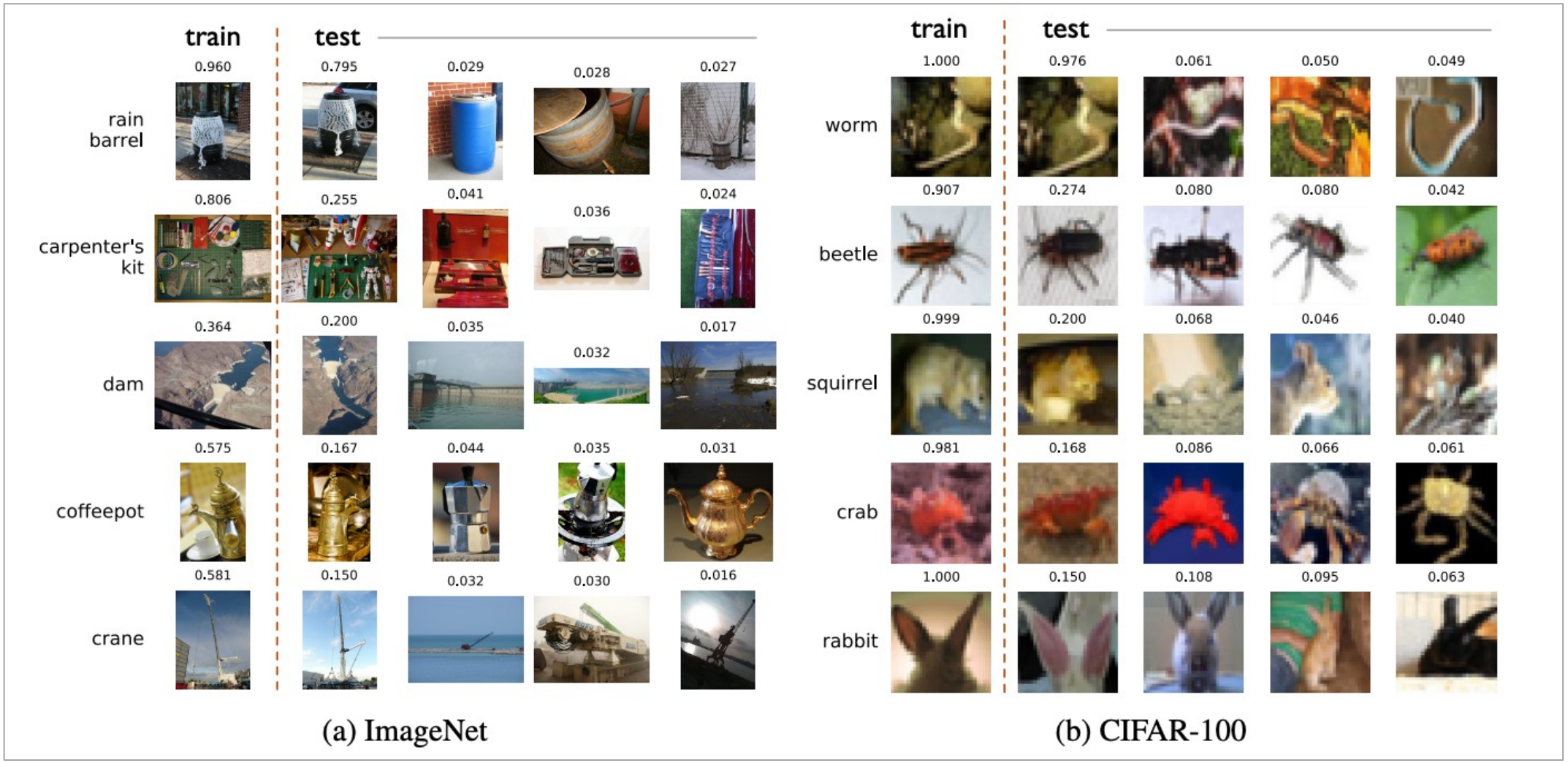
What Neural Networks Memorize and Why

Qualitative Examples where Memorization leads to Generalization



(a) ImageNet (b) CIFAR-100

What Neural Networks Memorize and Why

Summary

1. Memorization improves Generalization on real-world benchmarks (CIFAR-10, ImageNet).
2. This is because the distribution of real-world images are long-tailed.

Take Home Message

- **Generalization over Memorization**: DNNs tend to Generalize not Memorize.
- **Learning Process**: DNNS tend to first Generalize then Memorize.
- **Role of Gradient Descent**: The gradient descent provide implicit regularization for generalization.
- **Memorization and Visual Benchmarks**: Memorization can enhance generalization (=test performance) in standard visual benchmarks due to the long-tailed distribution of dataset.
- **Training with Random Labels**: Memorization significantly deteriorate the performance.

Recommended Readings

- Deep Learning through the lens of example difficulty., NeurIPS 2021.
  - Early layers tend to generalize while later layers tend to memorize.
- What Do Neural Networks Learn when trained with random labels., NeurIPS 2020.
  - Pre-training with random labels can bring positive transfer.
- Fortuitous Forgetting in Connectionist Networks., ICLR 2022.
  - Periodic Reinitialization of Last layers can improve Generalization.
- The Curious Case of Benign Memorization., ICLR 2023.
  - With Data Augmentation, Neural network can learn generalizable features from random labels.